

Dr. Mikel Sagardia

Principal Data Scientist
and Machine Learning Engineer
@ Deutsche Telekom

*I build full-stack, end-to-end AI applications in vision,
language, and 3D, working with teams to create real-
world impact.*

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location · San Sebastian

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publications · mikelsagardia.io/publications

Work Experience

Principal Data Scientist and ML Engineer @ Deutsche Telekom

Bonn (remote), 2023.05 — present

- Design and development of image segmentation and object detection applications
- Development of 3D processing algorithms for machine learning pipelines
- Research and development of vectorization algorithms for images and text
- Implementation of data clustering and downstream applications (pre-annotation, active learning, and search)
- Discovery of use cases for generative AI technologies
- Development, benchmarking and deployment of LLM-, RAG-, and agent-based solutions

Computer Vision and Machine Learning Engineer @ Self Employed

San Sebastian, 2022.04 — 2023.04 (1 year, 1 month)

- Built machine vision and machine learning applications as a contractor
- Supported and trained companies in the field of deep learning

Senior Researcher @ Vicomtech

San Sebastian, 2020.05 — 2021.12 (1 year, 8 months)

- Coordinated and executed European and national research projects
- Acquired new projects on computer vision, machine learning and robotics
- Supervised and mentored MSc and PhD students

Project Engineer in Computer Vision @ MVTec

Munich, 2019.03 — 2020.03 (1 year, 1 month)

- Managed and developed machine vision projects for key industry partners
- Trained customers and distributors on 3D vision and robotics
- Presented new features of the MVTec Halcon toolkit at the MVTec Innovation Day 2020
- Provided second level technical support in a broad spectrum of machine vision topics

Research Engineer @ German Aerospace Center (DLR)

Wessling / Munich, 2008.04 — 2018.12 (10 years, 9 months)

- Led a 5-year project that built a VR framework for satellite on-orbit servicing
- Researched into collision and force computation methods for haptic rendering
- Coordinated and executed cooperation projects for industry partners (VW, Bosch)
- Published of 22+ papers at IEEE Transactions on Haptics, IEEE ICRA, IEEE VR, and ACM VRST, among others
- Presented robotic applications on international trade fairs and congresses (Automatica, VW, etc.)
- Supervised and mentored 20+ students during their MSc theses and internships

Education

PhD in Electrical and Information Engineering @ Technische Universität München, TUM

Munich, 2013.12 — 2018.11

Dissertation: *Virtual Manipulations with Force Feedback in Complex Interaction Scenarios* (defended in 2019)

BSc & MSc in Mechanical Engineering @ Universidad de Navarra, Tecnun

San Sebastian, 2002.09 — 2008.02

Erasmus and Leonardo scholarships · Exchange student at *Technische Universität München*, TUM (Munich)

Skills

Machine Learning, CV & NLP: Pytorch & Lightning, Tensorflow & Keras, Scikit-Learn, Numpy, Pandas, SciPy, Matplotlib · OpenCV, Open3D, Halcon, PCL · NLTK, spaCy, HuggingFace Transformers & Diffusers, LangChain & LangGraph

Software Development: Python, C/C++, SQL, Bash · FastAPI, Flask, Streamlit, Gradio, HTML & CSS · Object Oriented Programming, Design Patterns, Data Structures, System Design

Cloud, MLOps & DevOps: AWS, Azure, PySpark, Elasticsearch & Kibana · Docker & Kubernetes, Git & CI/CD, MLflow, DVC, Pytest & Linting Tools

Languages (certificates available): English (advanced - C1), German (advanced - C1), Spanish (native), Basque (native)

Honors & Awards

Publication Awards: Best Video Award Finalist (IEEE ICRA 2014) · Best Paper Award (VAMR 2013) · Short Paper Honorable Mention (IEEE VR 2012)

Academic Awards: Second Best Diploma Thesis Prize 2008 (Association of Engineers of Gipuzkoa) · High School Excellence Award for University Access 2002 (Basque Government)